

DDTA research work plan after documentation-layer closure - R22

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WORK PLAN - REVISION 22

Status: active plan after BA5 canonical registry/controlled-authoring closure.

Supersedes: Revision 21 only for forward execution state; R1-R21 remain historical research records.

1. Fixed prior state

- Chapters 2-4: CLOSED / FINAL for current thesis scope.
- Documentation layer: CLOSED.
- BA0 responsibility/non-goals: CLOSED.
- BA1 minimal BAE identity ontology: CLOSED.
- BA2 relation/action vocabulary: CLOSED BY BA2-T4.
- BA3 provenance/derivation/identity/lifecycle/change-revalidation: CLOSED BY BA3-T4.
- BA4 projection boundary/traceability/coverage/interpretation: CLOSED BY BA4-T3.
- BA5 canonical semantic registry/controlled authoring: CLOSED BY BA5-T3.
- **BAReferent** and **BAProposition**: ACCEPTED first-class semantic identity families.
- ThreatForge remains a case study/tooling instrument, not DDTA semantic authority.

2. BA5 closure carried forward

BA5 closes the strict controlled-authoring hypothesis:

semantically operative DDTA binding
-> exact canonical token/name

unregistered alias / synonym
-> not normative semantic input

free explanatory prose
-> allowed outside semantic bindings

The closed registry responsibilities are domain-scoped: referent names, BA2 operators, operator-scoped roles, semantic kinds and controlled-value domains remain distinct in authority/scope even when physically co-located.

Normative synonym machinery is **REJECTED AS UNNECESSARY** for the current thesis scope. Optional lexical search, translation, migration or NLP/LLM assistance remains **DEFERRED** and must not become semantic authority.

3. BA5 extension/reopen rule

Do not reopen BA5 for convenience.

Reopen only the smallest BA5 responsibility on concrete evidence that exact canonical bindings plus presentation-local wording cannot preserve required shared meaning or reproducibility.

A materially new operator or role contract reopens BA2, not BA5. A new independently reusable identity applies the BA1 split criterion. Method-specific terminology remains downstream.

4. BA6 objective - complete Base Analysis integrated regression and closure

Status: NOT STARTED / NEXT PHASE.

BA6 is the integration/adversarial closure phase for the complete BA0-BA5 design. It is not a new ontology-building phase by default.

The integrated chain under test is:

```
governed documentation
-> canonical registry-controlled authoring
-> accepted Base Analysis
-> BA3 provenance / review / continuity
-> multiple BA4 projections
-> governed project change
-> BA revalidation / rebuild
-> projection rebuild / comparison
```

BA6 must determine whether the closed responsibilities work together without requiring consumers to reconstruct missing shared methodology-neutral meaning from raw prose.

5. BA6 evidence minimum

The phase must include:

1. regression over the closed facial-access corpus and mutation pressures;
2. regression over the closed order/WMS/provider corpus and change pressures;
3. at least one structurally different holdout not used to tune the BA0-BA5 lower bounds;
4. canonical-name and registry-revision resolution through the complete chain;
5. provenance/source drill-down from projection to BA to governed source;
6. change localization, BA3 continuity/revalidation and projection rebuild;
7. at least two materially different consumers/projections; and
8. explicit search for a material counterexample requiring the smallest BA0-BA5 reopen.

The holdout is for integration falsification, not for silently expanding DDTA into a universal systems-modeling language.

6. BA6 guardrails

Do not:

- add a new BAE family without a material BA1 counterexample;
- add a BA2 operator/role merely for convenience;
- weaken BA3 source/provenance/change accountability;
- let a projection repair missing shared BA meaning by rereading raw prose;
- let method taxonomy redefine BA semantics;
- reintroduce synonym normalization without a BA5 material counterexample;
- treat ThreatForge runtime classes as DDTA semantics;
- start AnalysisRecord, Common Finding or formal STRIDE/STRIDE-AI schema design; or
- declare Base Analysis closed before the integrated regression passes.

7. BA6 counterexample routing

Use the smallest-responsibility rule:

```
material counterexample
-> identify exact failed responsibility
-> reopen only that BA phase/contract
-> repair and rerun affected regression
```

Examples:

- missing independent semantic identity -> BA1;
- missing method-neutral proposition/operator/role semantics -> BA2;
- unreproducible provenance/change continuity -> BA3;
- projection coverage/trace/interpretation failure -> BA4;

- canonical registry/authoring failure -> BA5.

Do not redesign Base Analysis wholesale from one local failure.

8. BA6 exit condition

BA6 may close the complete Base Analysis milestone only if:

1. the closed corpora regress without material semantic loss;
2. the structurally different holdout does not force an unresolved shared-semantic gap;
3. source drill-down and historical reproducibility remain intact;
4. governed change can be localized and revalidated without whole-BA invalidation by default;
5. multiple projections can be rebuilt without becoming project authority;
6. canonical registry/controlled-authoring semantics survive integration; and
7. any discovered counterexample has been routed to and resolved in the smallest responsible phase.

Only then may the state become:

BASE ANALYSIS CLOSED FOR CURRENT THESIS SCOPE

9. Downstream analysis envelope remains blocked

Until BA6 closes, do not start:

- A1 - AnalysisRecord / execution envelope;
- A2 - common reviewed Finding boundary;
- A3 - analysis change/provenance integration;
- formal STRIDE/STRIDE-AI overlay schemas; or
- ThreatForge implementation work that would imply unclosed semantic authority.

10. Next authorized work

The next authorized phase is:

BA6 - complete Base Analysis integrated regression and closure.

Before execution, define the smallest first BA6 microstep against the closed BA0-BA5 contract. Revision 22 intentionally does not invent a BA6 microstep count or execute BA6 inside the BA5-T3 closure package.